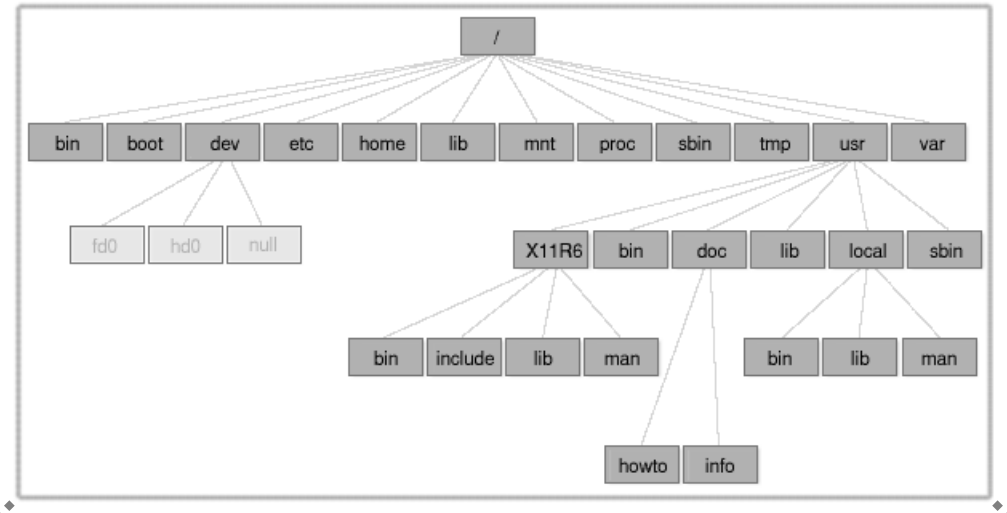
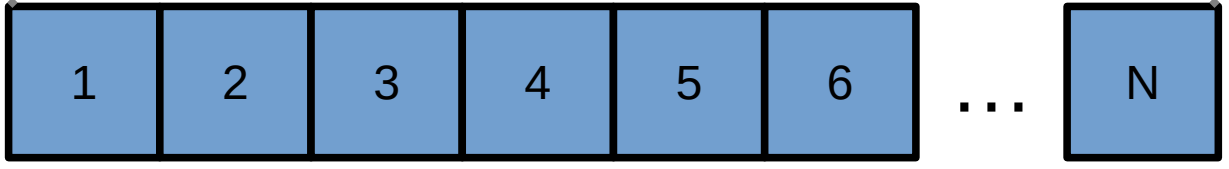
Файловая система

Назначение файловой системы (ФС)

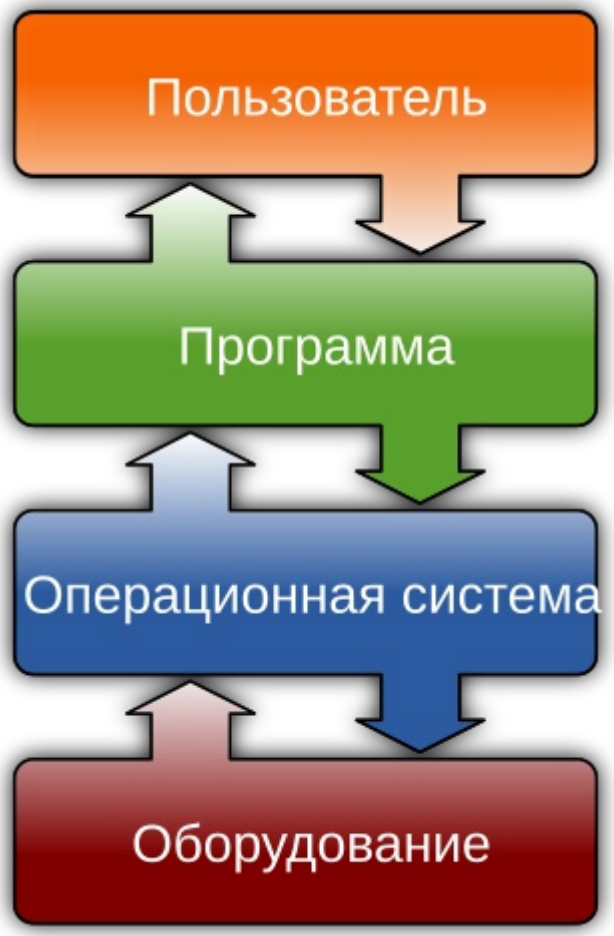
Данные:



Байты:

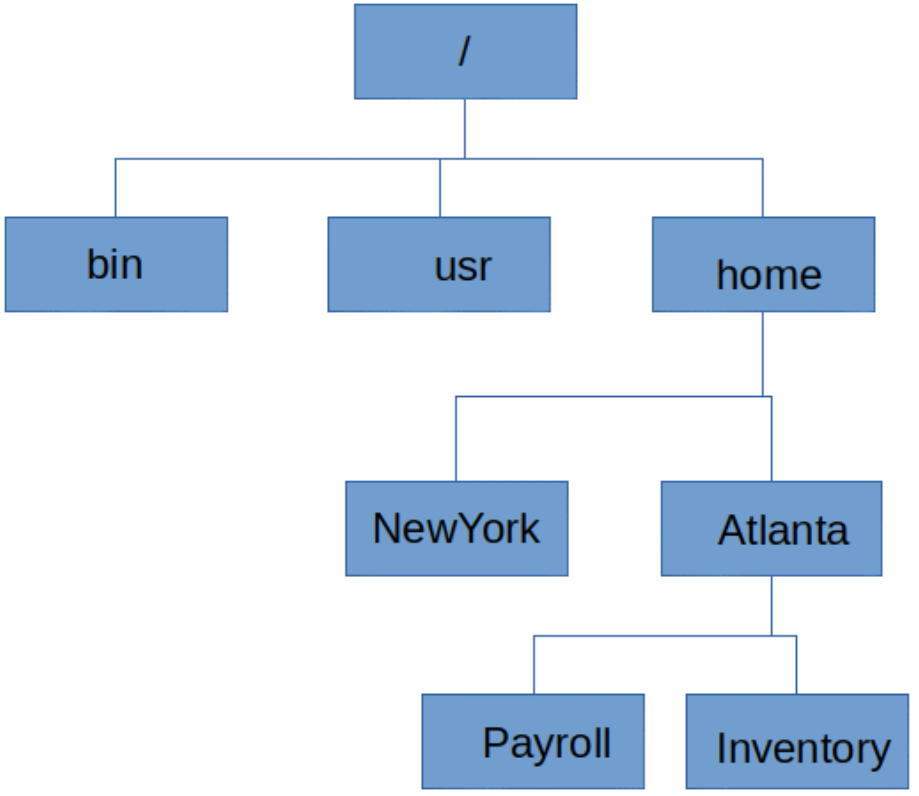


Storage



Классификация ФС — 1

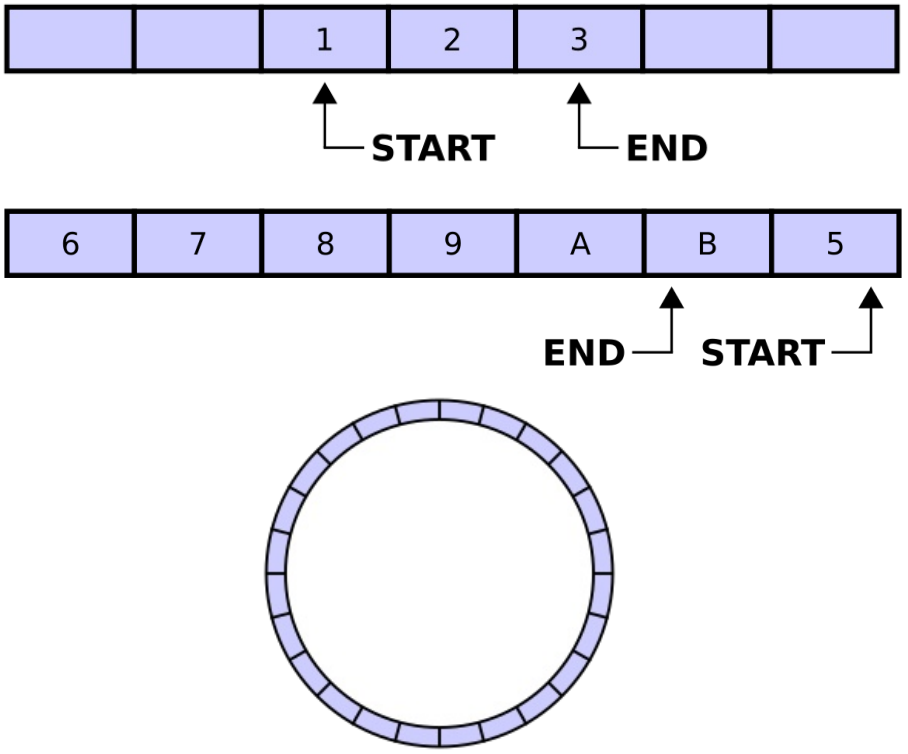
- Одноуровневые
  - MS-DOS 1.0, встраиваемые системы
- Иерархические (1965)
  - MULTICS, UNIX, GNU/Linux, ...



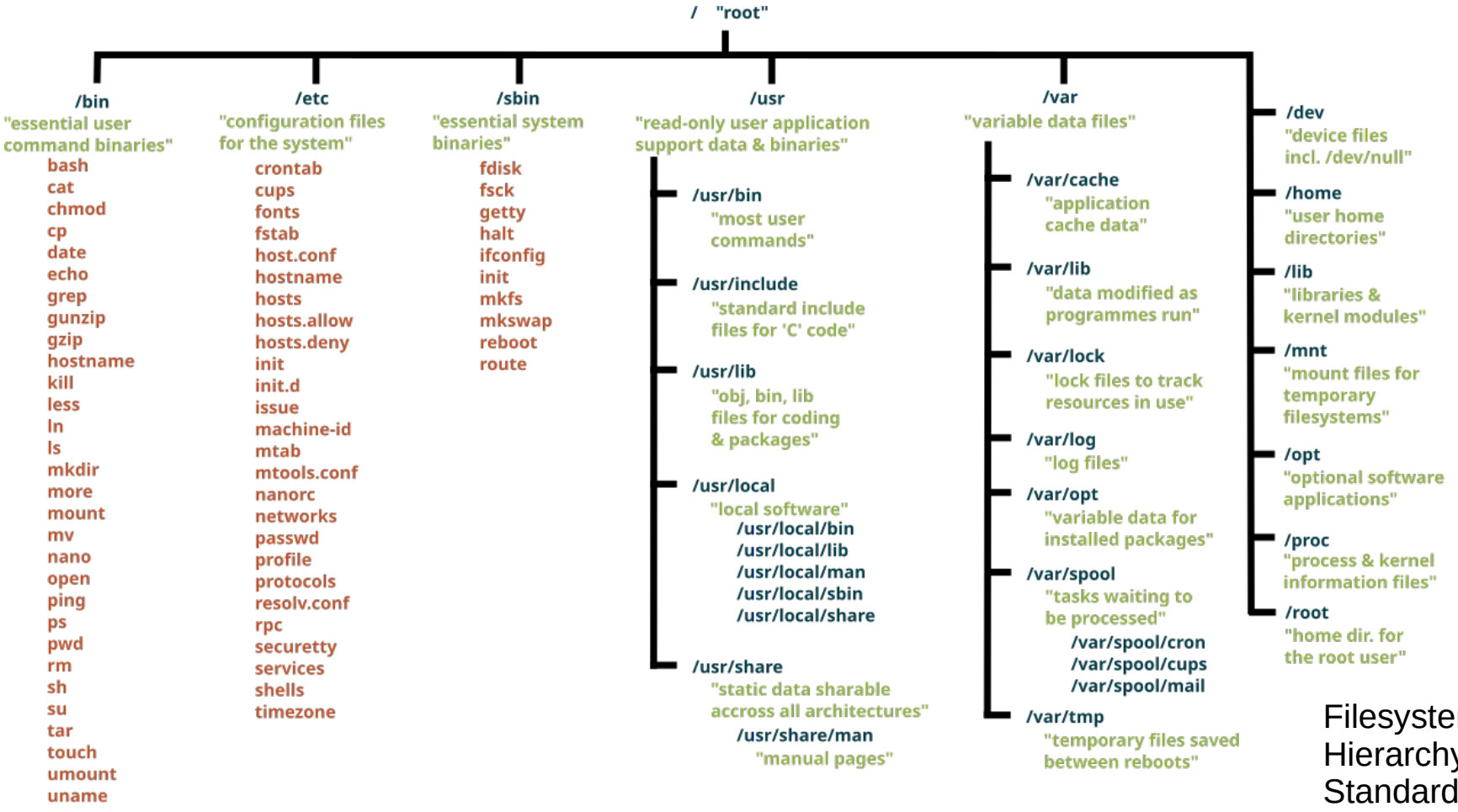
Классификация ФС — 2

- Журналируемые
  - JFS, HFS Plus, APFS, NTFS, ext3, ext4
- Не-журналируемые
  - fat, ext, ext2

Циклический буфер:



Стандарт файловой системы

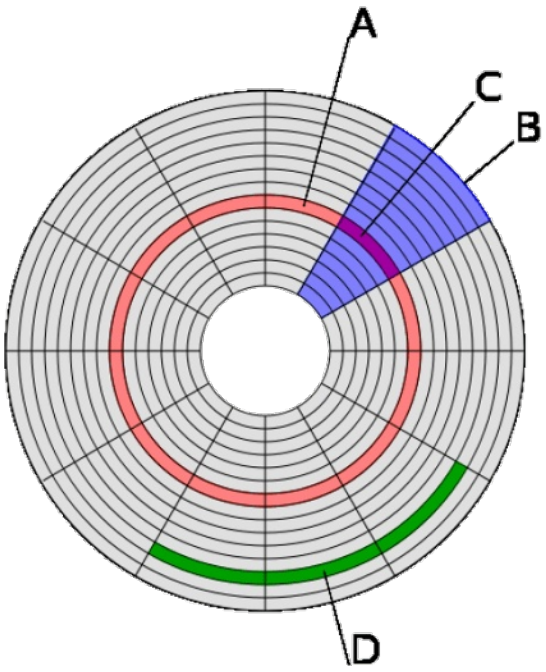


Filesystem Hierarchy Standard (FHS)

Физический накопитель

```
$ fdisk -l
[…]
Disk /dev/sdb: 465,76 GiB,
500107862016 bytes, 976773168
sectors
Disk model: Samsung SSD 860
Units: sectors of 1 * 512 = 512
bytes
Sector size (logical/physical): 512
bytes / 512 bytes
I/O size (minimum/optimal): 512
bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x8d302f21

[...]
```



Hard Drive Structure:

A = track
B = sector
C = sector of a track
D = cluster

https://wiki.archlinux.org/title/Advanced_Format

Системный вызов «stat»

```
int stat(const char* path, struct stat* buf);
int lstat(const char* path, struct stat* buf);
int fstat(int filedesc, struct stat* buf);
```

```
#include <sys/stat.h>

struct stat {
    dev_t     st_dev;     /* ID of device containing file */
    ino_t     st_ino;     /* Inode number */
    mode_t    st_mode;    /* File type and mode */
    nlink_t   st_nlink;   /* Number of hard links */
    uid_t     st_uid;     /* User ID of owner */
    gid_t     st_gid;     /* Group ID of owner */
    dev_t     st_rdev;    /* Device ID (if special file) */
    off_t     st_size;    /* Total size, in bytes */
    blksize_t st_blksize; /* Block size for filesystem I/O */
    blkcnt_t  st_blocks;  /* Number of 512 B blocks allocated */

    /* Since POSIX.1-2008, this structure supports nanosecond
       precision for the following timestamp fields.
       For the details before POSIX.1-2008, see HISTORY.  */

    struct timespec  st_atim;  /* Time of last access */
    struct timespec  st_mtim;  /* Time of last modification */
    struct timespec  st_ctim;  /* Time of last status change */

#define st_atime  st_atim.tv_sec  /* Backward compatibility */
#define st_mtime  st_mtim.tv_sec
#define st_ctime  st_ctim.tv_sec
};
```

280 блоков * 512 байт = 143360 байт

Предпочитаемый размер выделения памяти

```
$ stat /usr/bin/ls
  File: /usr/bin/ls
  Size: 142312     Blocks: 280        IO Block: 4096   regular file
Device: 252,1  Inode: 45099330    Links: 1
Access: (0755/-rwxr-xr-x)  Uid: (    0/    root)   Gid: (    0/    root)
Access: 2026-09-25 17:48:47.461612872 +0300
Modify: 2026-08-25 18:09:03.000000000 +0300
Change: 2026-09-01 08:49:50.430151050 +0300
 Birth: 2026-09-01 08:49:50.357150433 +0300
```

https://docs.kernel.org/filesystems/ext4/blocks.html

Просмотр информации о ФС

```
struct statvfs {
    unsigned long  f_bsize;    /* Filesystem block size */
    unsigned long  f_frsize;   /* Fragment size */
    fsblkcnt_t     f_blocks;   /* Size of fs in f_frsize units */
    fsblkcnt_t     f_bfree;    /* Number of free blocks */
    fsblkcnt_t     f_bavail;   /* Number of free blocks for
                                  unprivileged users */
    fsfilcnt_t     f_files;    /* Number of inodes */
    fsfilcnt_t     f_ffree;    /* Number of free inodes */
    fsfilcnt_t     f_favail;   /* Number of free inodes for
                                  unprivileged users */
    unsigned long  f_fsid;     /* Filesystem ID */
    unsigned long  f_flag;     /* Mount flags */
    unsigned long  f_namemax;  /* Maximum filename length */
};
```

$ **df** -hT | grep root

/dev/mapper/root ext4 314G 247G 53G 83% /

$ **stat** --file-system /bin/ls

File: "/bin/ls"

ID: d261a64b31bce02f Namelen: 255 Type: ext2/ext3

Block size: 4096 Fundamental block size: 4096

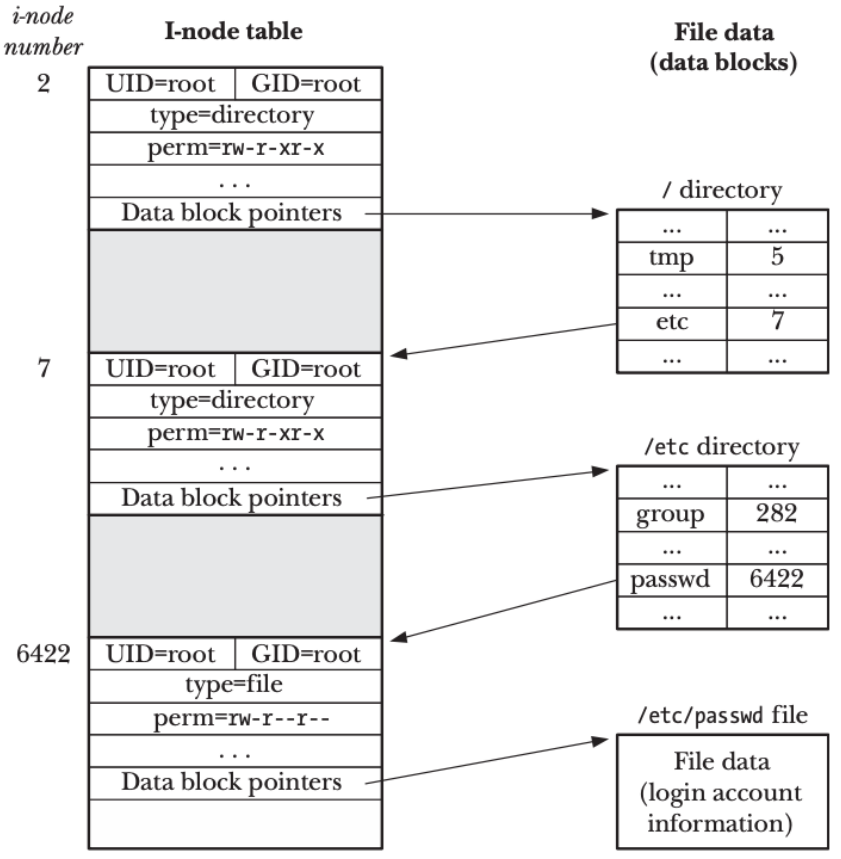
Blocks: Total: 82306031 Free: 17760290 Available: 13792145

Inodes: Total: 20979712 Free: 16467443

82306031 blocks * 4096 bytes ≈ 313 GB

Index node (inode)

- Хранит метаданные:
  - Device ID
  - File mode
  - Link count
  - User/Group ID
  - Size
  - Timestamps
  - Block number
  - …
- Номера с 0 по 10 зарезервированы
- 32 бита / 64 бита

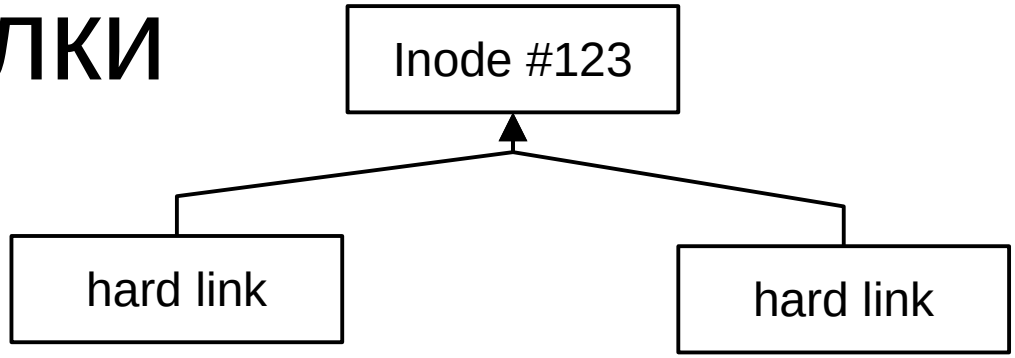


Информация об inode

```
$ df -i
Filesystem          Inodes   IUsed   IFree      IUse% Mounted on
[...]
/dev/mapper/root   20979712 4512923 16466789   22%   /
[...]

$ ls -i file.txt
221 file.txt
```

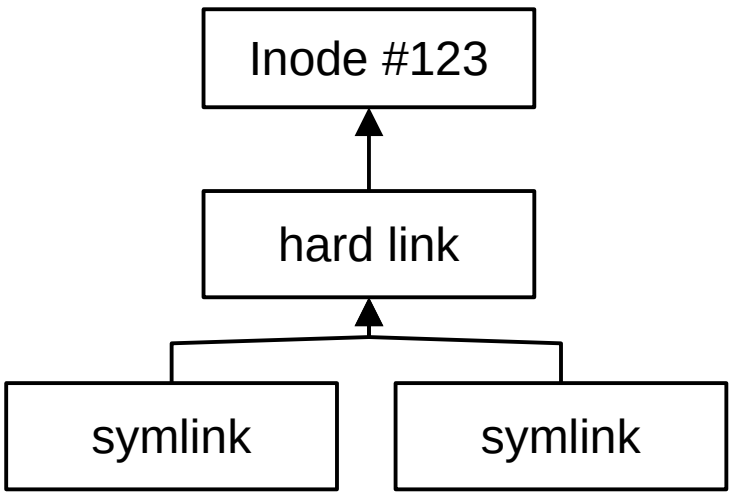
Ссылки



- Виды ссылок
  - Жёсткие ссылки
  - Символические (symlink)

Создание жёсткой ссылки:

```
$ ln file1 file2
```



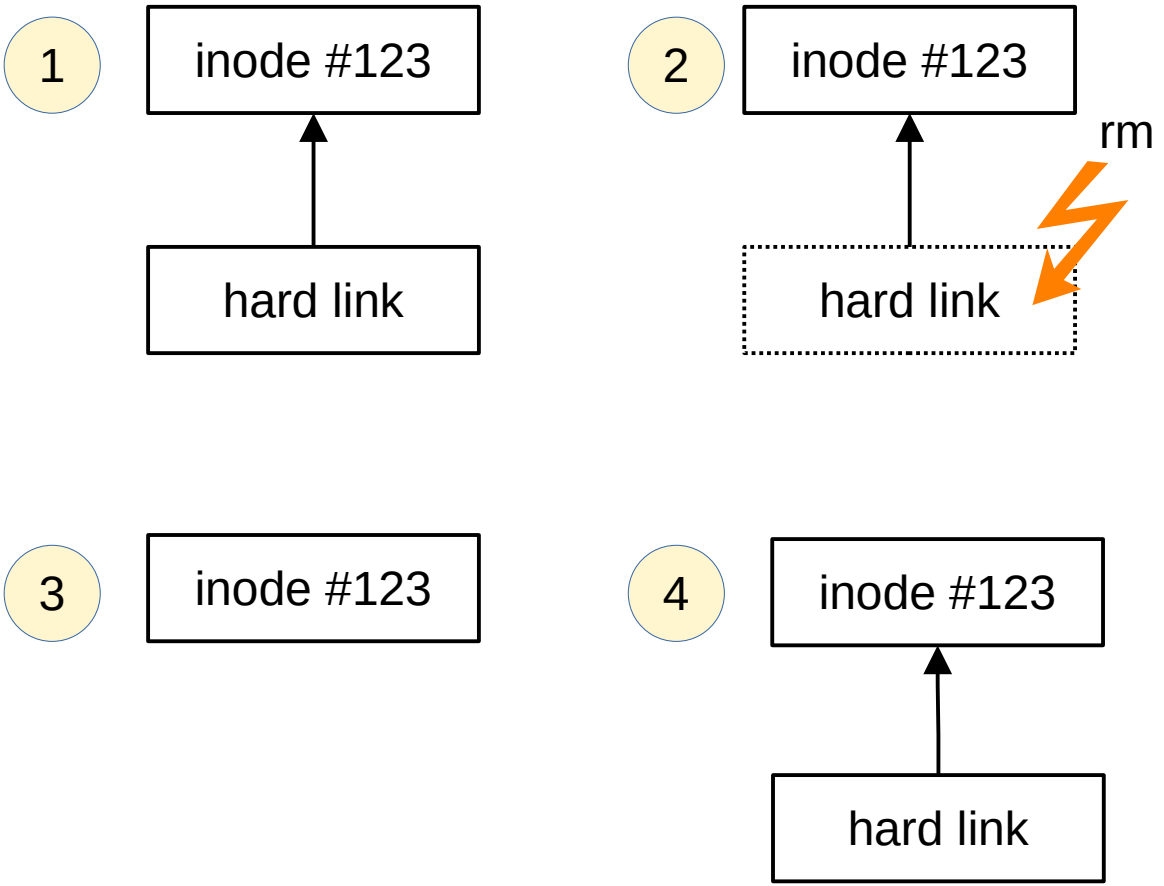
Создание симлинка:

```
$ ln -s file3 file4
```

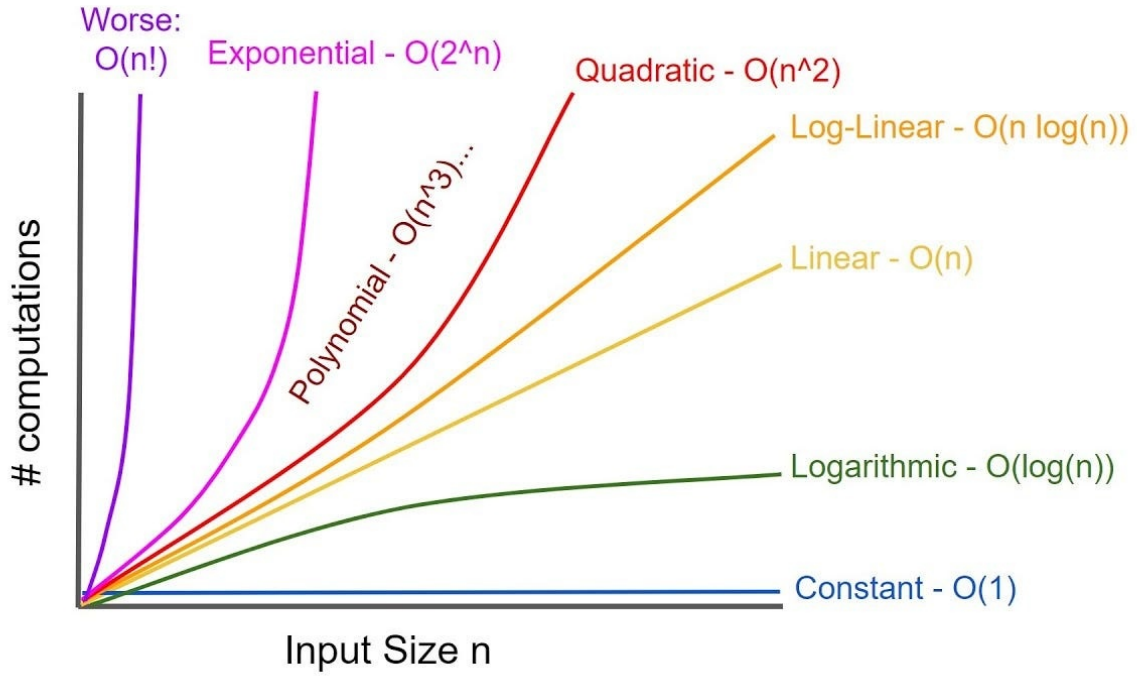
Удаление файла

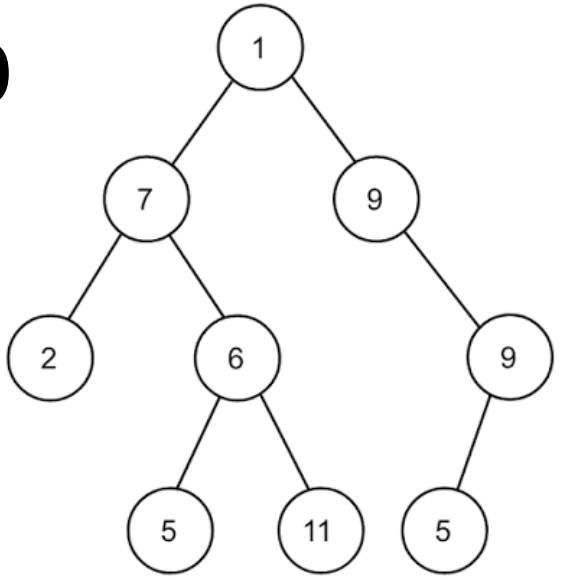
$ touch file.txt

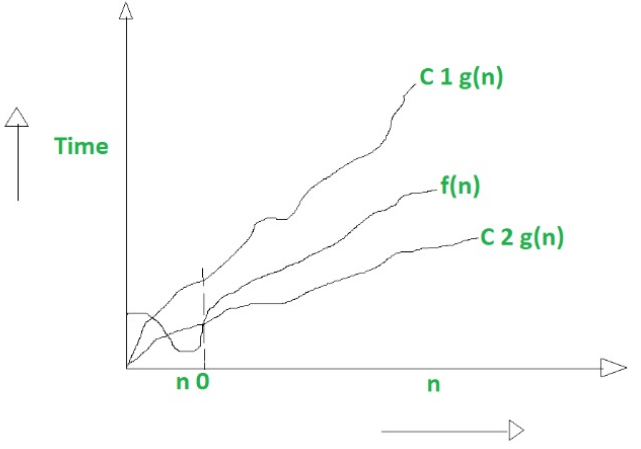
$ rm file.txt



Бинарное дерево

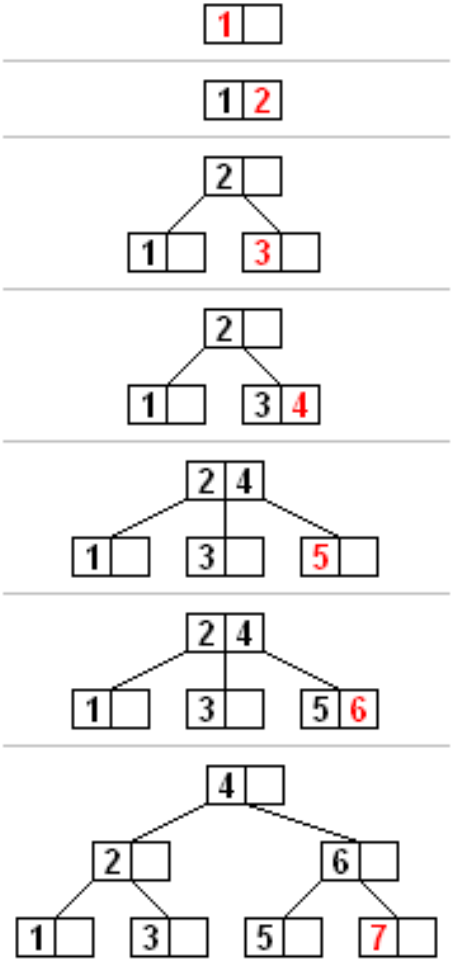






B-tree

- Сбалансированное дерево поиска, в котором каждый узел содержит множество ключей и имеет более двух потомков.
- Порядок дерева «m»
  - Количество потомков: m
  - Максимальное количество ключей: m - 1
  - Минимальное количество ключей: m / 2



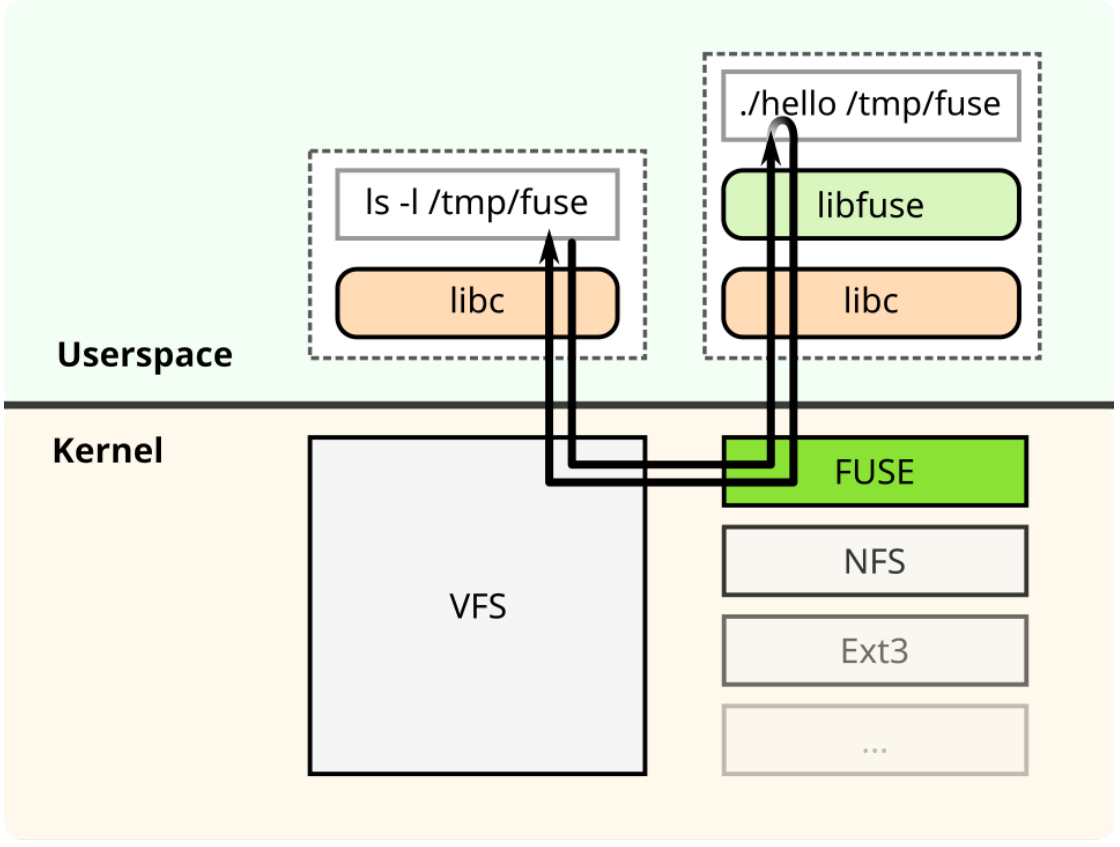
Специальные ФС

- /proc
- /dev
- /sys

```
$ mount | grep -E ' on /(proc|sys|dev)(/| )'
sysfs on /sys type sysfs (rw,nosuid,nodev,noexec,relatime)
proc on /proc type proc (rw,nosuid,nodev,noexec,relatime)
udev on /dev type devtmpfs (rw,nosuid,relatime,size=65826476k,nr_inodes=16456619,mode=755,inode64)
devpts on /dev/pts type devpts (rw,nosuid,noexec,relatime,gid=5,mode=620,ptmxmode=000)
securityfs on /sys/kernel/security type securityfs (rw,nosuid,nodev,noexec,relatime)
tmpfs on /dev/shm type tmpfs (rw,nosuid,nodev,inode64)
cgroup2 on /sys/fs/cgroup type cgroup2 (rw,nosuid,nodev,noexec,relatime,nsdelegate,memory_recursiveprot)
pstore on /sys/fs/pstore type pstore (rw,nosuid,nodev,noexec,relatime)
efivarfs on /sys/firmware/efi/efivars type efivarfs (rw,nosuid,nodev,noexec,relatime)
bpf on /sys/fs/bpf type bpf (rw,nosuid,nodev,noexec,relatime,mode=700)
systemd-1 on /proc/sys/fs/binfmt_misc type autofs (rw,relatime,fd=32,pgrp=1,timeout=0,minproto=5,maxproto=5,direct,pipe_ino=27541)
mqueue on /dev/mqueue type mqueue (rw,nosuid,nodev,noexec,relatime)
debugfs on /sys/kernel/debug type debugfs (rw,nosuid,nodev,noexec,relatime)
hugetlbfs on /dev/hugepages type hugetlbfs (rw,nosuid,nodev,relatime,pagesize=2M)
tracefs on /sys/kernel/tracing type tracefs (rw,nosuid,nodev,noexec,relatime)
fusectl on /sys/fs/fuse/connections type fusectl (rw,nosuid,nodev,noexec,relatime)
configfs on /sys/kernel/config type configfs (rw,nosuid,nodev,noexec,relatime)
binfmt_misc on /proc/sys/fs/binfmt_misc type binfmt_misc (rw,nosuid,nodev,noexec,relatime)
```

Filesystem in Userspace (FUSE)

- Примеры:
  - SSHFS
  - NTFS-3G
  - ZFS



Источники изображений

- Hard drive tracks / sectors / clusters illustration © Heron2/MistWiz — Public Domain: https://commons.wikimedia.org/wiki/File:Disk-structure2.svg

Sven

- FUSE / libfuse / VFS diagram © Sven — GNU FDL 1.2: https://commons.wikimedia.org/wiki/File:FUSE_structure.svg